MA 465: Introduction to Mathematical Modeling

Project 5

Difference Equations and National Income Model

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1. Abstract

This laboratory investigates the use of second-order difference equations in economic modeling, focusing on the Samuelson (1939) model for national income. The general solution of the difference equation is derived and analyzed in terms of the coefficients A, B, and C, highlighting their influence on system behaviors such as convergence, divergence, and oscillation. Specific parameter sets are simulated and plotted to illustrate these dynamics, including cases from the original Samuelson paper. Additionally, a special case demonstrates the equivalence between the Samuelson model and the classical compound interest problem, emphasizing the connection between discrete-time economic models and fundamental financial growth principles.

2. Introduction

This laboratory examines the behavior of second-order difference equations and their application to economic modeling, specifically the Samuelson (1939) model for national income. The focus is on understanding the general solution of a second-order difference equation, relating its coefficients to the parameters of the Samuelson model, and analyzing the resulting dynamic behaviors.

The general solution of the second-order equation is derived and its behaviors such as oscillation, convergence, or divergence—is discussed in terms of the coefficients (A,B,C) and their relationship to the Samuelson model. Solutions are then computed and plotted for specific parameter sets, including the cases reported in Fig. 5.4 of the textbook and selected regions from the original Samuelson paper, highlighting how different parameter values produce distinct system behaviors. Finally, a special case where the difference equation reduces to the compound interest problem is examined, demonstrating the connection between economic dynamics and classical financial mathematics.

3. Methodology

First, a general solution of the Samuelson model was obtained. This solution was divided into three cases, each analyzed and described in terms of its behavior. Using this solution, the coefficients (A, B, C) were expressed in terms of the model parameters (α, β, γ).

Next, MATLAB was used to plot the three cases illustrated in Figure 5.4 of the course textbook. In these plots, coefficients B and C were held constant while A was varied so that each plot corresponded to a different case of the general solution.

Subsequently, MATLAB was employed to generate plots for the three regions (B, C, and D) discussed in Samuelson's original paper, in order to identify which solution case each region corresponds to.

Finally, the solution with coefficients A=4 and B=C=0 was compared to the compound interest problem with p=A−1 and r=1, demonstrating that the two problems are equivalent under this parameter set.

4. General Solution of the Samuelson Model

The Samuelson Model for national income is governed by the following equations and parameters:

$$y_n = C_n + I_n + G_n \tag{1}$$

$$\begin{cases} C_n = \alpha y_{n-1} \\ I_n = \beta(C_n - C_{n-1}) \\ G_n = \gamma \end{cases} \tag{2}$$

where C_n represents the total household spending on goods and services, I_n represents private investment responding to changes in consumption, and G_n represents fixed government spending. The parameters α and β reflect the reactions of the model to past income and changes in consumption, respectively.

The Samuelson model is a second-order difference equation of the type:

$$y_n = Ay_{n-1} + By_{n-2} + C \tag{3}$$

From equation 2, the investment term can be rewritten as

$$I_n = \beta(C_n - C_{n-1}) = \beta(\alpha y_{n-1} - \alpha y_{n-2}) = \alpha\beta(y_{n-1} - y_{n-2}) \tag{4}$$

Equations 1, 2, 3, and 4 can then be combined to express the Samuelson model explicitly as:

$$y_n = C_n + I_n + G_n$$

$$y_n = \alpha y_{n-1} + \alpha\beta(y_{n-1} - y_{n-2}) + \gamma$$

$$y_n = \alpha(1+\beta)y_{n-1} - \alpha\beta y_{n-2} + \gamma \tag{5}$$

From equations 3 and 5, the relationship between (A, B, C) and (α, β, γ) can be easily determined as:

$$\begin{cases} A = \alpha(1+\beta) \\ B = -\alpha\beta \\ C = \gamma \end{cases} \tag{6}$$

To find the general solution of equation 5 we start with the homogeneous portion of the second-order difference equation in equation 3 and assume a solution of the form $y_n^h = r^n$:

$$y_n^h = Ay_{n-1} + By_{n-2}$$

$$y_n^h - Ay_{n-1} - By_{n-2} = 0$$

$$r^n - Ar^{n-1} - Br^{n-2} = 0 \tag{7}$$

By assuming $r \neq 0$ and dividing every term in equation 7 by r^{n-2} the characteristic equation of the solution:

$$r^2 - Ar - B = 0$$

$$r_{1,2} = \frac{A \pm \sqrt{A^2 + 4B}}{2} = \frac{A}{2} \pm \sqrt{\frac{A^2}{4} + B} \tag{8}$$

To obtain a particular solution, a term satisfying $s = As + Bs + C$ must be added to the homogeneous solution:

$$s = As + Bs + C$$

$$s = \frac{C}{1 - A - B} \tag{9}$$

Thus, the general solution for the Samuelson model of national income is

$$y_n = k_1 r_1^n + k_2 r_2^n + s = k_1 r_1^n + k_2 r_2^n + \frac{C}{1 - A - B} \tag{10}$$

where k_1 and k_2 can be determined by initial conditions y_0 and y_1 and the solution falls into one of three cases

$$\begin{cases} y_n = k_1 r_1^n + k_2 r_2^n \; for\ distinct\ real\ roots & (11a) \\ y_n = (k_1 + k_2 n) r^n \; for\ repeated\ real\ roots & (11b) \\ y_n = p^n [k_1 \cos(n\theta) + k_2 \sin(n\theta)] \; for\ complex\ roots & (11c) \end{cases}$$

For the case in equation 11a, the solution is unbonded if $|r_{1,2}| > 1$. If $|r_{1,2}| < 1$ the solution tends to the value of s:

$$\lim_{n \to \infty} y_n = s = \frac{C}{1 - A - B}$$

For the case in equation 11b, the solution is unbounded if $\left|\frac{A}{2}\right| > 1$. If $\left|\frac{A}{2}\right| < 1$, the solution tends to the value of s, as in the case of equation 11a:

$$\lim_{n \to \infty} y_n = s = \frac{C}{1 - A - B}$$

For the case of equation 11c, the solution is a dampening oscillation if $|k| < 1$ and an amplified oscillation if $|k| > 1$.

4. Data

After deriving the general solution, three specific cases of the Samuelson model were considered with specific formulas, given in the course textbook, for A and B. These formulas and the varying coefficients B_0 are summarized below and in Table 1.

$$\begin{cases} A = \dfrac{2}{B_0} - \sqrt{\dfrac{4}{B_0}\left(\dfrac{C}{s} - 1 + \dfrac{1}{B_0}\right)} \\ B = -\dfrac{B_0}{4}A^2 \\ C = 0.1 \end{cases}$$

Case	B_0
1	0.1
2	1
3	1.05

Table 1: B_0 Values

These conditions were used to generate three plots in MATLAB, shown in Figure 1 below.

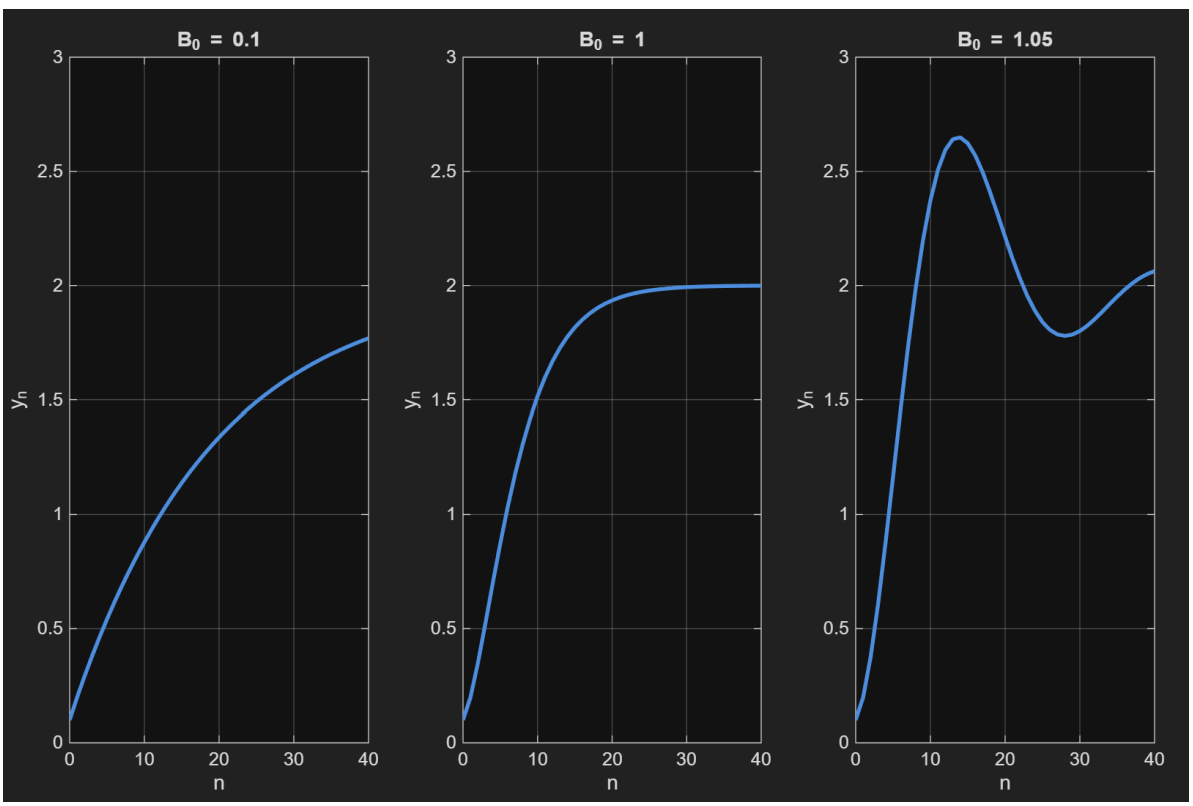

Figure 1: y_n vs. n (years)

Next, regions B, C, and D given in the Samuelson paper (1939) were graphed in MATLAB. To obtain each region's plot, the coefficient (α, β, γ) values were varied according to Table 2. The resulting graphs can be found in Figure 2.

Region	$(\boldsymbol{\gamma}, \boldsymbol{\alpha}, \boldsymbol{\beta})$
B	(1, 0.5, 1)
C	(1, 0.6, 2)
D	(1, 0.8, 4)

Table 2: Regions According to Coefficients (α, β, γ)

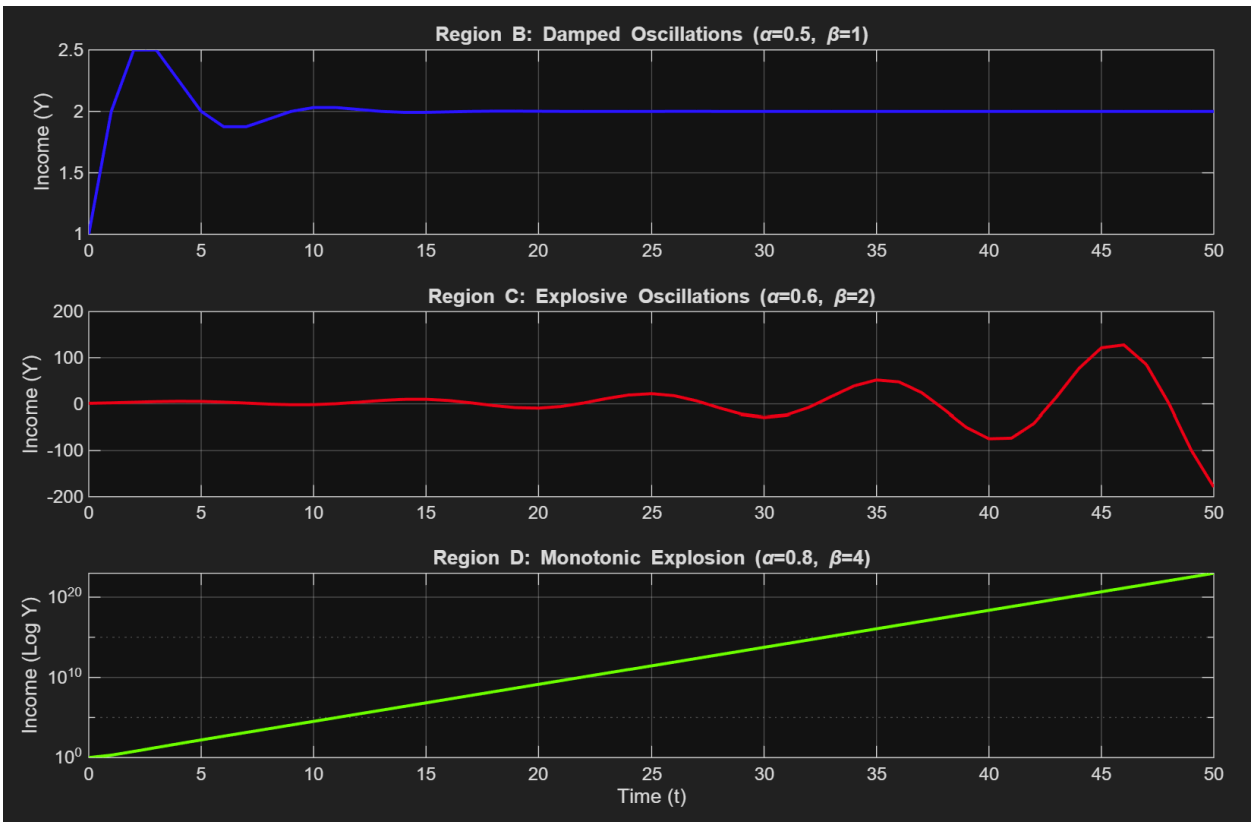

Figure 2: Regions

Finally, the Samuelson model was compared to the compound interest problem. To obtain a reasonable comparison, the Samuelson coefficients and compound interest parameters were set according to Table 3. Figure 3 shows the resulting graph.

Coefficient/Parameter	**Value**
A	4
B	0
C	0
p	A-1=3
r	1

Table 3: Samuelson vs Compound Interest Values

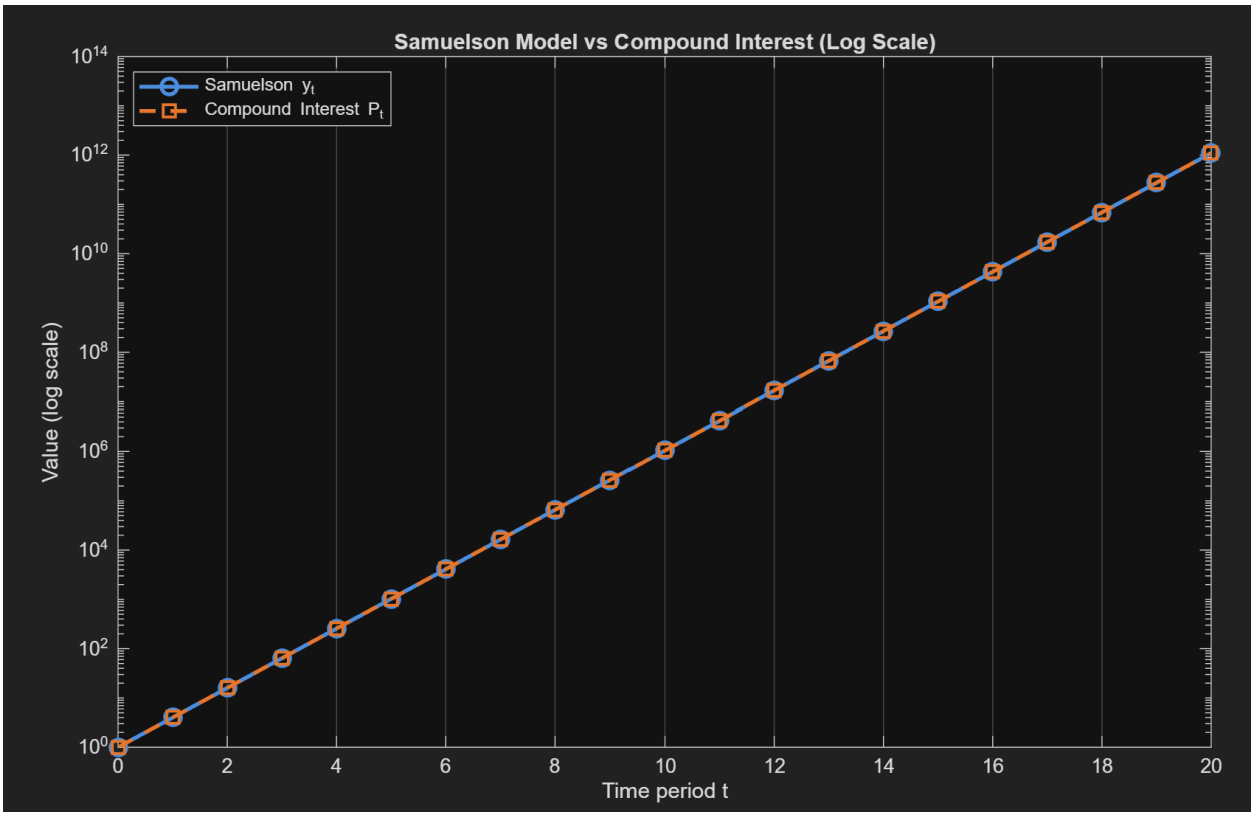

Figure 3: Samuelson Model vs. Compound Interest

5. Discussion

Figure 1 shows three cases of the Samuelson model. Each of the three cases is defined by its B_0 value. All three values converge to $y_n = 2$. The first plot with a B_0 value of 0.1 shows a solution with non-repeated real roots, meaning no oscillations occur before the value converges. The second plot with a B_0 value of 1 shows a solution with repeated real roots, meaning the system is critically dampened. The third plot shows a solution with complex conjugate roots, meaning the system is a damped oscillation. The value of the third plot will continue to oscillate around $y_n = 2$ before converging.

Figure 2 shows three of the regions discussed in the Samuelson paper. The behavior of the national income model in these three cases is determined by the value of the discriminant of the model's characteristic equation. Region B and Region C both belong to the complex roots solution, which occurs when the discriminant is negative, triggering the oscillatory cycles seen in the first two plots in Figure 2. The distinction between Regions B and C occurs due to a second factor, the value of the product $\alpha\beta$. In Region B, the product is less than one, leading to damped oscillations that converge. In Region C, the product exceeds one, causing explosive oscillations that grow in amplitude. Conversely, Region D belongs to the distinct real roots solution because its parameters result in a positive discriminant. In this case, the absence of complex components eliminates all oscillatory behavior, and

since the magnitude of at least one root is greater than 1, the income model continues growing towards infinity.

Figure 3 shows a comparison between specific instances of the Samuelson model and the compound interest problem. The values for both models perfectly overlap, as expected.

6. Conclusion

This report demonstrates how second-order difference equations can effectively model dynamic economic systems, as exemplified by the Samuelson model for national income. By deriving the general solution and analyzing the characteristic roots, we showed how the coefficients A, B, and C determine whether the system exhibits convergence, divergence, or oscillatory behavior. Graphical simulations for different parameter sets confirmed the theoretical predictions, illustrating the range of possible economic dynamics, from smooth stabilization to cyclical fluctuations. The special case connecting the Samuelson model to the compound interest problem further highlights the fundamental link between discrete-time economic models and classical financial growth, emphasizing the broader applicability of difference equations in understanding both economic and financial processes.

7. Appendices

Appendix A: Complete MATLAB code for graphing and calculations:

```
% Part B
clear; clc;

% Parameters
C = 0.1;
s = 2;
B0_vec = [0.1, 1, 1.05];

t_end = 40;
n = 0:t_end;

figure;

for i = 1:length(B0_vec)

   B0 = B0_vec(i);

   % Compute A using Eq. (5.74)
   A = (2/B0) - sqrt((4/B0)*(C/s - 1 + 1/B0));

   % Compute B
```

```
    B = -(B0/4)*(A^2);

    % Initialize y
    y = zeros(1, length(n));

    % Initial conditions
    y(1) = 0.1;   % y_0
    y(2) = 0.2;   % y_1

    % Difference equation
    for k = 3:length(n)
        y(k) = A*y(k-1) + B*y(k-2) + C;
    end

    % Plot
    subplot(1,3,i)
    plot(n, y, 'LineWidth', 2)
    grid on

    title(['B_0 = ', num2str(B0)])
    xlabel('n')
    ylabel('y_n')

    ylim([0 3])
    xlim([0 40])
end
% ---------------------------------------------------------------------
% Part C
t_max = 50; % Number of time periods
t = 0:t_max;

% Define Parameters
alpha_B = 0.5; beta_B = 1;
alpha_C = 0.6; beta_C = 2;
alpha_D = 0.8; beta_D = 4;

figure;

% --- Plot Region B (Linear Scale) ---
subplot(3,1,1);
plot(t, run_samuelson(alpha_B, beta_B, t_max), 'b', 'LineWidth', 1.5);
title('Region B: Damped Oscillations (\alpha=0.5, \beta=1)');
ylabel('Income (Y)');
grid on;
```

```
% --- Plot Region C (Linear Scale) ---
subplot(3,1,2);
plot(t, run_samuelson(alpha_C, beta_C, t_max), 'r', 'LineWidth', 1.5);
title('Region C: Explosive Oscillations (\alpha=0.6, \beta=2)');
ylabel('Income (Y)');
grid on;

% --- Plot Region D (Logarithmic Scale) ---
subplot(3,1,3);
% Use semilogy to make the y-axis powers of 10
semilogy(t, run_samuelson(alpha_D, beta_D, t_max), 'g', 'LineWidth', 1.5);
title('Region D: Monotonic Explosion (Log Scale: Powers of 10) (\alpha=0.8, \beta=4)');
ylabel('Income (Log Y)');
xlabel('Time (t)');
grid on;

% Difference Equation Function
function Y = run_samuelson(a, b, t_max)
   Y = zeros(1, t_max + 1);
   Y(1) = 1; % Setting Y_0 to 1 to avoid log(0) issues in Region D
   Y(2) = 2;
   for t = 3:t_max+1
      Y(t) = a*(1+b)*Y(t-1) - a*b*Y(t-2) + 1;
   end
end

% ----------------------------------------------------------------------
% Part D
clc; clear; close all;

% Parameters
A = 4;      % Samuelson coefficient
B = 0;      % Samuelson coefficient
C = 0;      % Samuelson constant term
y0 = 1;     % Initial value for Samuelson
P0 = 1;     % Initial principal for compound interest
t_max = 20;  % Number of periods (longer time)
t = 0:t_max;

% Samuelson model solution (B=C=0)
y = y0 * A.^t;

% Compound interest solution with p = A-1
```

```
p = A - 1;          % Interest rate
P = P0 * (1 + p).^t;  % Compound interest

% Plotting (logarithmic y-scale)
figure;
semilogy(t, y, 'o-', 'LineWidth', 2, 'MarkerSize', 8);  % Samuelson
hold on;
semilogy(t, P, 's--', 'LineWidth', 2, 'MarkerSize', 8); % Compound interest

xlabel('Time period t');
ylabel('Value (log scale)');
title('Samuelson Model vs Compound Interest (Log Scale)');
legend('Samuelson y_t', 'Compound Interest P_t', 'Location', 'northwest');
```